Preface

ADCS – Spacecraft Attitude Determination and Control was written to provide an accessible book on designing spacecraft control systems. The idea was to mix theory and practice so that the reader can immediately implement the concepts discussed in this book. For that reason, MATLAB® code is given for most key elements in the book. The code can easily be converted to Python or other languages at the preference of the reader.

ADCS is a huge field. The first control systems were implemented in the 1960s prior to the introduction of digital flight computers. As a consequence, there is a vast literature on the subject. In this book, I have tried to cover the full range of approaches, from passive control all the way to Artificial Intelligence-based approaches. In this way, the reader will not have to follow in the pattern of designing a new system like the last system they did, just because it is familiar. I have tried to be compact in the presentation so you will not find long derivations or theorems and proofs. In some cases just enough information is provided to give you a starting point for further research. Most of the book is based on my experiences in spacecraft control. I do discuss other spacecraft for which there are good references in the literature.

I have been working in the area of spacecraft control for forty years, starting with learning about the Space Shuttle Orbiter at the Charles Stark Draper Laboratory, even writing code in HAL/S the Space Shuttle programming language, through working on numerous control-systems designs at GE Astro Space, and then working on many new concepts at my company, Princeton Satellite Systems. Back when I started Princeton Satellite Systems, there were only a couple of dozen engineers who had designed spacecraft control systems. Nowadays, starting with the introduction of CubeSats through the massive commercial constellations, there are hundreds, if not thousands of experienced engineers. Even those with experience will hopefully find this book valuable, if only to provide other approaches to the problems they need to solve. That said, the approaches in this book are not the only way to handle the problems and the reader should not feel constrained in any way by the ideas presented in the book.

Part 1 of the book outlines ADCS theory. Many short examples are given to show how the theory works in practice. Classes of spacecraft are presented to motivate the theory and give the reader a way to quickly use the material. The book tries to make chapters self-contained so the reader will see some repetition of material. This is intentional.

Part 2 of the book gives examples from industry. Examples range from CubeSats to solar sails. The book does not necessarily replicate the control systems that are actually used, instead it shows the process to design an ADCS system that meets the general requirements of the mission.

The appendices present background theory in control, statistics, math, and other topics that are helpful to understand the material in the book.

The book does not feature any problems at the end of the chapters. Instead, it provides MATLAB source code for all of the examples. Once you have set up your paths to the supporting function folder, you can run all of the code. The code includes short snippets to explain a point in the text and complete simulations. The provided code can be used as a starting point for your work. For experienced ACS designers, the book may provide alternative approaches to ACS design. For other subsystem designers, it gives a thorough overview of ACS design so that you will know what the ACS people are doing.

This book was written over the course of many years. Some of the material is in the reference book for my company's software products. I have incorporated material that I use in teaching an MIT Attitude Control course. The newest material, on Artificial Intelligence, has its origins in 6.034 Artificial Intelligence taught by Professor Patrick Winston at MIT in 1973, and various explorations in AI on different spacecraft projects.

I hope you enjoy the book. Please contact me with your questions, suggestions, or corrections!

Michael Paluszek